

Infrastructure Sizing Details

ROSHN Parametric Masterplan Modelling Platform

Version: 1.0 **Date:** 5 January 2026 **Status:** Production Ready **Purpose:** Infrastructure provisioning guide for ROSHN IT

Note: The application stack is certified production-ready. Any future enhancements or modifications based on ROSHN feedback will be implemented as iterative improvements and do not affect the current production readiness status.

Executive Summary

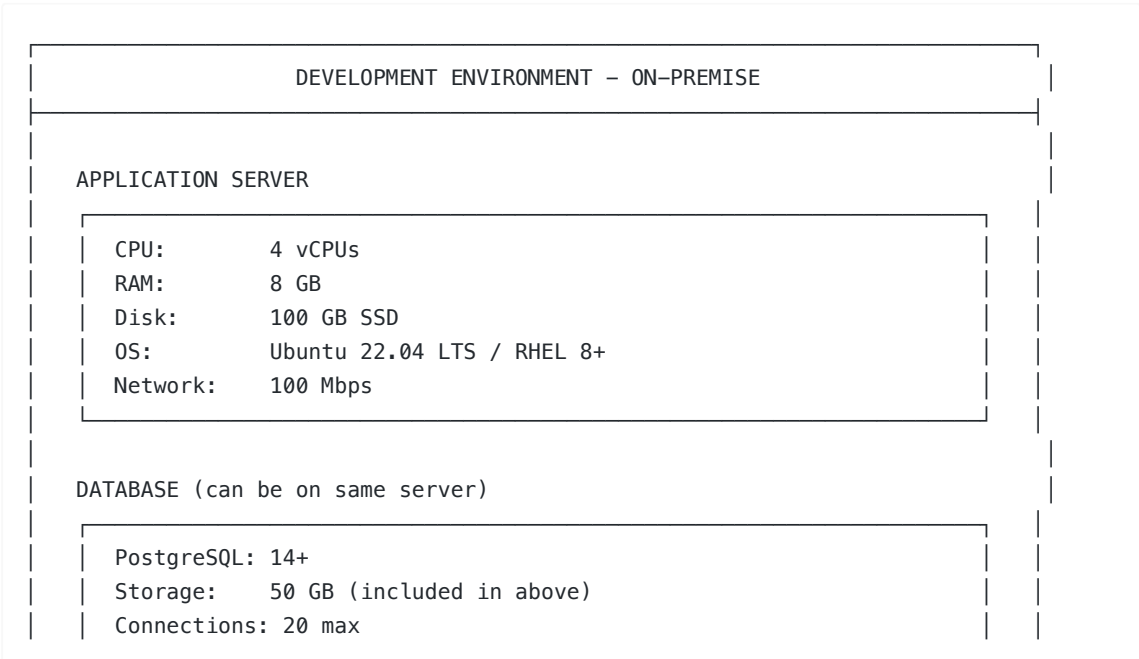
This document provides detailed infrastructure specifications for deploying the ROSHN Platform across three environments (Development, Staging, Production) and three deployment options (Azure, GCP, On-Premise).

Environment Overview

Environment	Purpose	Availability	Users
Development	Feature development and testing	Business hours	5-10 developers
Staging	UAT and pre-production testing	Business hours	10-20 testers
Production	Live business operations	24/7	50-100+ users

1. Development Environment

1.1 On-Premise Specifications



Component	Specification
CPU	4 vCPUs (Intel Xeon or equivalent)
RAM	8 GB DDR4
Storage	100 GB SSD (NVMe preferred)
Operating System	Ubuntu 22.04 LTS or RHEL 8+
Network	100 Mbps dedicated
Database	PostgreSQL 14+ (same server)
SSL	Optional (self-signed acceptable)

1.2 Azure Specifications

Resource	SKU	Specification
App Service Plan	B1 (Basic)	1 core, 1.75 GB RAM
App Service	Linux	Node.js 20 LTS
PostgreSQL	B_Gen5_1	1 vCore, 2 GB RAM, 32 GB storage
Storage Account	Standard LRS	10 GB for file uploads

Azure Resource Group: roshn-platform-dev-rg

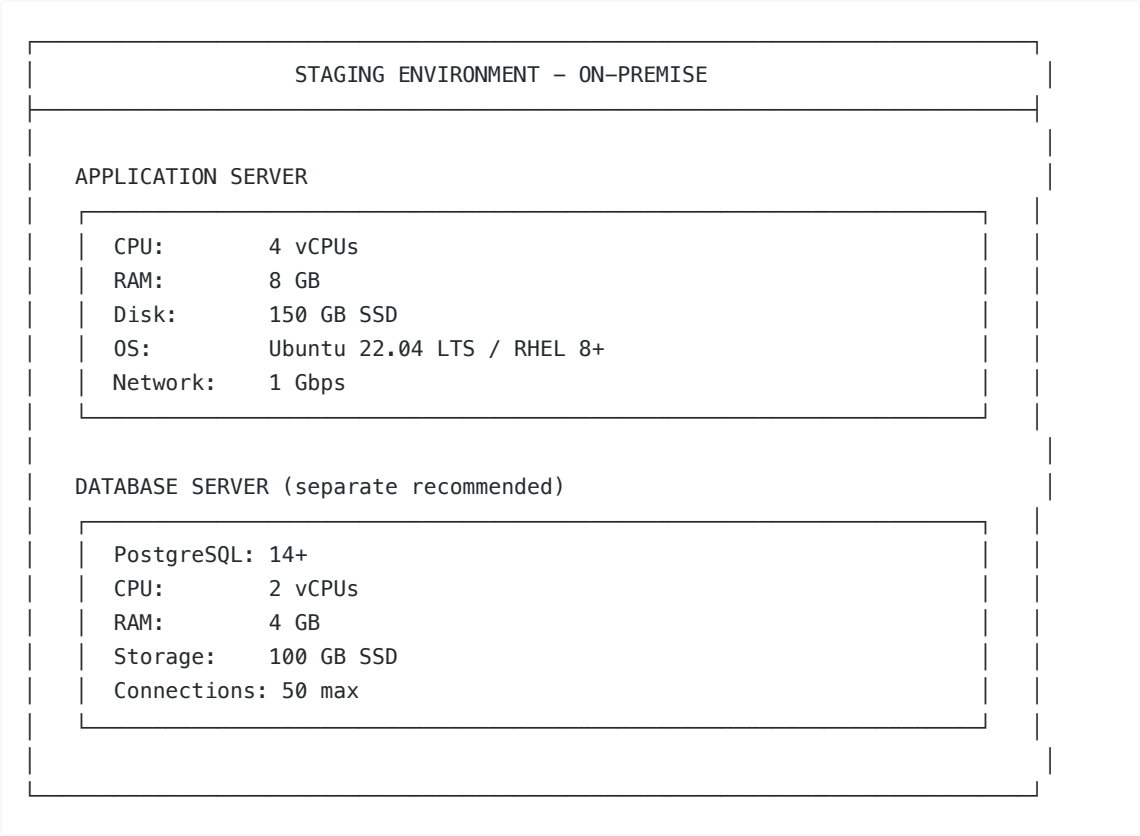
1.3 GCP Specifications

Resource	Type	Specification
Cloud Run	-	512 Mi RAM, 1 vCPU
Cloud SQL	db-f1-micro	Shared core, 0.6 GB RAM
Cloud Storage	Standard	10 GB for file uploads
Min Instances	0	Scale to zero when idle
Max Instances	2	Limited scaling

GCP Project: roshn-platform-dev

2. Staging Environment

2.1 On-Premise Specifications



Component	Specification
CPU	4 vCPUs (Intel Xeon or equivalent)
RAM	8 GB DDR4
Storage	150 GB SSD (NVMe preferred)
Operating System	Ubuntu 22.04 LTS or RHEL 8+
Network	1 Gbps dedicated
Database	PostgreSQL 14+ (separate server recommended)
SSL	Required (valid certificate)
Load Balancer	Optional

2.2 Azure Specifications

Resource	SKU	Specification
App Service Plan	S1 (Standard)	1 core, 1.75 GB RAM
App Service	Linux	Node.js 20 LTS
PostgreSQL	GP_Gen5_2	2 vCores, 4 GB RAM, 128 GB storage
Storage Account	Standard LRS	50 GB for file uploads

Azure Resource Group: roshn-platform-staging-rg

2.3 GCP Specifications

Resource	Type	Specification
Cloud Run	-	1 Gi RAM, 1 vCPU
Cloud SQL	db-custom-1-3840	1 vCPU, 3.75 GB RAM
Cloud Storage	Standard	50 GB for file uploads
Min Instances	0	Scale to zero when idle
Max Instances	5	Moderate scaling

GCP Project: roshn-platform-staging

3. Production Environment

3.1 On-Premise Specifications

PRODUCTION ENVIRONMENT – ON-PREMISE

APPLICATION SERVERS (2x for HA)

CPU: 8 vCPUs (each)
RAM: 16 GB (each)
Disk: 250 GB SSD (each)
OS: Ubuntu 22.04 LTS / RHEL 8+
Network: 1 Gbps

DATABASE SERVER (dedicated)

PostgreSQL: 14+
CPU: 4 vCPUs
RAM: 16 GB
Storage: 500 GB SSD (RAID 10)
Connections: 200 max
Replication: Standby recommended

LOAD BALANCER

Type: L7 Load Balancer (Nginx/HAProxy)
SSL: Termination at LB
Health: HTTP health checks

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Component	Specification
Application Servers	2x servers for high availability
CPU (per server)	8 vCPUs (Intel Xeon or equivalent)
RAM (per server)	16 GB DDR4 ECC
Storage (per server)	250 GB SSD (NVMe, RAID 1)
Operating System	Ubuntu 22.04 LTS or RHEL 8+
Network	1 Gbps dedicated, redundant
Database Server	Dedicated server
Database CPU	4 vCPUs
Database RAM	16 GB DDR4 ECC
Database Storage	500 GB SSD (RAID 10)
SSL	Required (valid certificate)
Load Balancer	Required
Backup Storage	1 TB for backups

3.2 Azure Specifications

Resource	SKU	Specification
App Service Plan	P1V2 (Premium)	1 core, 3.5 GB RAM
App Service	Linux	Node.js 20 LTS, 2 instances min
PostgreSQL	GP_Gen5_4	4 vCores, 8 GB RAM, 256 GB storage
Storage Account	Standard GRS	100 GB for file uploads
Azure CDN	Standard	Static asset delivery
Application Insights	-	Monitoring and logging

Azure Resource Group: roshn-platform-rg

3.3 GCP Specifications

Resource	Type	Specification
Cloud Run	-	2 Gi RAM, 2 vCPU
Cloud SQL	db-custom-2-4096	2 vCPU, 4 GB RAM

Cloud Storage	Standard	100 GB for file uploads
Cloud CDN	-	Static asset delivery
Min Instances	1	Always running
Max Instances	10	Auto-scaling
Cloud Monitoring	-	Logging and alerts

GCP Project: roshn-platform

4. Comparison Summary

4.1 Resource Comparison by Environment

Resource	Development	Staging	Production
App CPU	4 vCPU	4 vCPU	8 vCPU × 2
App RAM	8 GB	8 GB	16 GB × 2
App Storage	100 GB	150 GB	250 GB × 2
DB CPU	Shared	2 vCPU	4 vCPU
DB RAM	2 GB	4 GB	16 GB
DB Storage	50 GB	100 GB	500 GB
Network	100 Mbps	1 Gbps	1 Gbps × 2
Load Balancer	No	Optional	Yes
SSL	Optional	Yes	Yes
HA	No	No	Yes

4.2 Cloud SKU Comparison

Environment	Azure App	Azure DB	GCP Run	GCP SQL
Development	B1	B_Gen5_1	512Mi/1CPU	db-f1-micro
Staging	S1	GP_Gen5_2	1Gi/1CPU	db-custom-1-3840
Production	P1V2	GP_Gen5_4	2Gi/2CPU	db-custom-2-4096

5. Network Requirements

5.1 Required Ports

Port	Protocol	Service	Access
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22	TCP	SSH	Internal only (management)
80	TCP	HTTP	Public (redirects to 443)
443	TCP	HTTPS	Public
3000	TCP	Application	Internal only
5432	TCP	PostgreSQL	Internal only

5.2 Firewall Rules

FIREWALL CONFIGURATION

INBOUND RULES

Allow: TCP/443 from 0.0.0.0/0 (HTTPS)
Allow: TCP/80 from 0.0.0.0/0 (HTTP redirect)
Allow: TCP/22 from VPN/Management subnet only
Deny: All other inbound

INTERNAL RULES

Allow: TCP/5432 from App servers to DB server
Allow: TCP/3000 from Load balancer to App servers

OUTBOUND RULES

Allow: TCP/443 to Azure AD (SSO)
Allow: TCP/443 to Mapbox API
Allow: TCP/443 to npm registry (updates)

5.3 Bandwidth Requirements

Environment	Minimum	Recommended
Development	100 Mbps	100 Mbps
Staging	100 Mbps	1 Gbps
Production	1 Gbps	1 Gbps (redundant)

6. Storage Requirements

6.1 Database Storage Estimates

Data Type	Est. Size (Year 1)	Est. Size (Year 3)
Users	< 1 MB	< 5 MB
Masterplans	~50 MB	~500 MB
Benchmark Projects	~100 MB	~1 GB
Cost Model Data	~20 MB	~100 MB
Audit Logs	~200 MB	~2 GB
Total	~400 MB	~4 GB

6.2 File Storage Estimates

File Type	Est. Size (Year 1)	Est. Size (Year 3)
CSV Uploads	~100 MB	~500 MB
Documents	~500 MB	~2 GB
Images	~200 MB	~1 GB
Total	~800 MB	~3.5 GB

6.3 Backup Storage

Environment	Backup Retention	Storage Required
Development	7 days	10 GB
Staging	14 days	50 GB
Production	30 days	200 GB

7. Software Requirements

7.1 Required Software Stack

Software	Version	Purpose
Node.js	20 LTS	Application runtime
npm	10+	Package management
PostgreSQL	14+	Database
Nginx	Latest	Reverse proxy
PM2	Latest	Process manager
Git	Latest	Version control

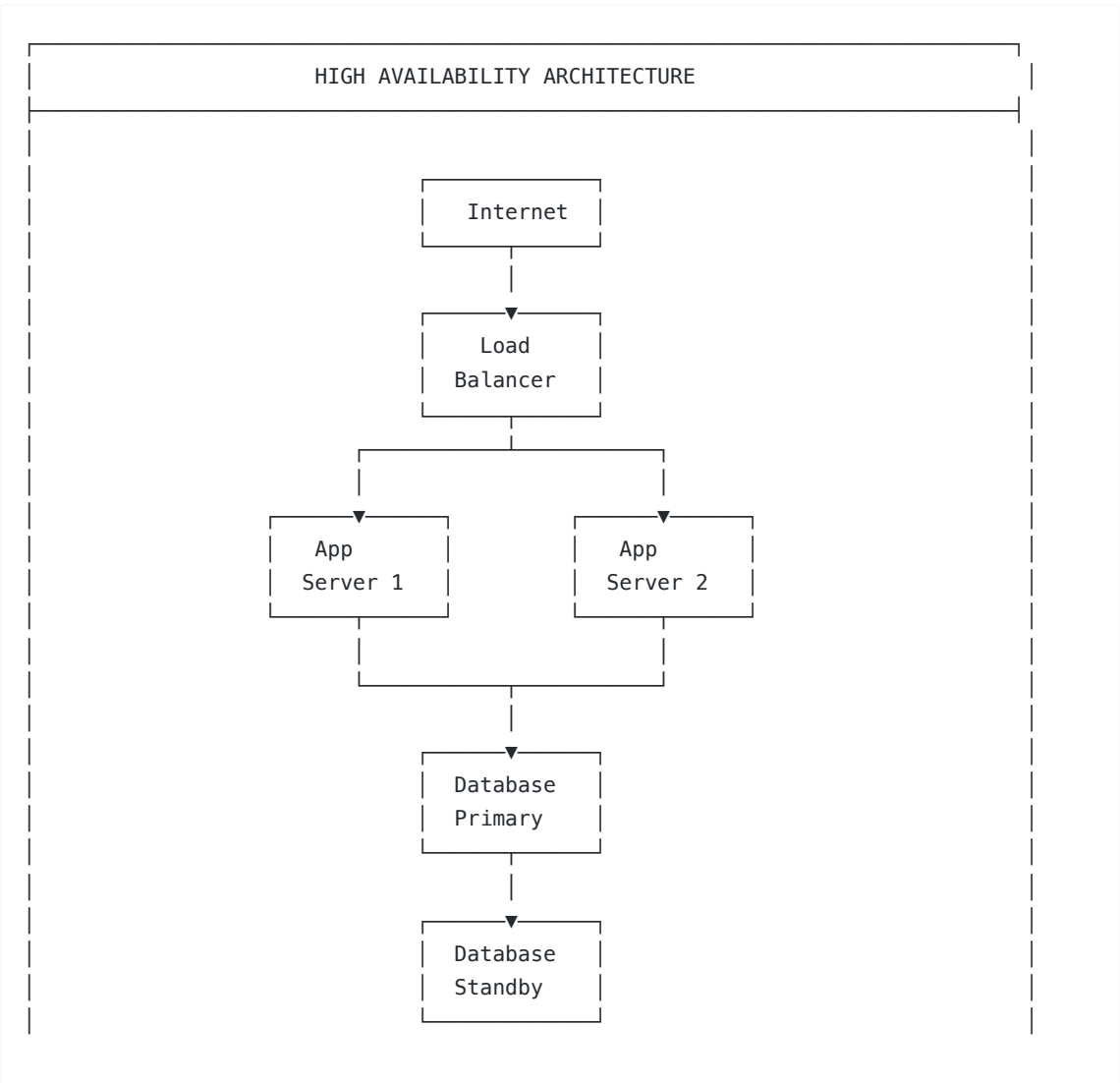
OpenSSL	Latest	SSL/TLS
Certbot	Latest	SSL certificate management

7.2 Optional Software

Software	Purpose
Redis	Session caching (future)
Elasticsearch	Log aggregation (future)
Prometheus	Metrics collection
Grafana	Monitoring dashboard

8. High Availability Configuration

8.1 Production HA Architecture



8.2 Failover Configuration

Component	Failover Method	RTO
Application	Load balancer auto-failover	< 30 seconds
Database	Streaming replication	< 5 minutes
Load Balancer	Floating IP (if applicable)	< 1 minute

9. Monitoring Requirements

9.1 Metrics to Monitor

Category	Metrics
Application	Response time, error rate, requests/sec
Database	Connections, query time, disk usage
System	CPU, RAM, disk I/O, network
Business	Active users, masterplans created

9.2 Alerting Thresholds

Metric	Warning	Critical
CPU Usage	> 70%	> 90%
Memory Usage	> 75%	> 90%
Disk Usage	> 70%	> 85%
Response Time	> 2s	> 5s
Error Rate	> 1%	> 5%
DB Connections	> 70%	> 90%

10. Provisioning Checklist

10.1 Development Environment

- ☐ Provision application server (4 vCPU, 8 GB RAM, 100 GB SSD)
- ☐ Install Ubuntu 22.04 LTS or RHEL 8+
- ☐ Install Node.js 20 LTS
- ☐ Install PostgreSQL 14+
- ☐ Configure firewall rules

- ☐ Create database and user
- ☐ Deploy application

10.2 Staging Environment

- ☐ Provision application server (4 vCPU, 8 GB RAM, 150 GB SSD)
- ☐ Provision database server (2 vCPU, 4 GB RAM, 100 GB SSD)
- ☐ Install operating systems
- ☐ Install required software
- ☐ Configure SSL certificate
- ☐ Configure firewall rules
- ☐ Setup monitoring
- ☐ Deploy application

10.3 Production Environment

- ☐ Provision 2x application servers (8 vCPU, 16 GB RAM, 250 GB SSD each)
- ☐ Provision database server (4 vCPU, 16 GB RAM, 500 GB SSD)
- ☐ Provision standby database server
- ☐ Configure load balancer
- ☐ Install operating systems
- ☐ Install required software
- ☐ Configure SSL certificate
- ☐ Configure firewall rules
- ☐ Setup backup solution
- ☐ Setup monitoring and alerting
- ☐ Configure database replication
- ☐ Deploy application
- ☐ Test failover procedures

11. Contact Information

For infrastructure provisioning questions, contact:

Role	Contact
Technical Lead	Gavin Britton
Operations	Timothy Shelton
ROSHN IT	Viquar Hashmi